# Foodhub Data Analysis

## Project Python Foundations

05/18/2025

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# Executive Summary

Conclusions

- The FoodHub is generating a lot of food orders by individuals and families but there are no large catering orders for parties or events. The majority of the orders are low cost.
- The most favorite cuisine is American food. There are a few other cuisines like Japanese, Italian and Chinese.  Majority of the orders come from these big Four!
- Weekends is the popular time to order food. Weekends generate many more orders than weekdays. The food delivery times on weekends are better than weekdays possibly even with the extra traffic.
- There are also a few favorite restaurants that are the big revenue generators. Four restaurants could be identified as generating top revenue and having excellent ratings.
- There is a lot of scope to correct the ratings collection for the app. There are almost 39% of orders that don’t have any ratings.
- There does not seem to be any correlation between food delivery and preparation time and cost of the food.
- Most of the customers order food only once.

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# Executive summary

Actionable Insights

- There are four favorite cuisines that generate major amount of revenue. Further analysis may be needed as to why other cuisines are not getting that much revenue.
- When new restaurants join the FoodHub app, focus should be on the most popular cuisines that are the big revenue generators. This can boost the business.
- On the weekdays, the delivery time can be reduced by deploying more delivery persons in the areas where the orders are high. This can reduce the delivery time and can boost the user experience.
- There is a need to generate a survey to understand why the customers are not rating the orders and why they are not ordering food many times. Promotional offers can be given to the customers to review and rate the orders.

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# Executive summary

Recommendations

- The user ratings collection system is very weak. The database column needs to be converted to a number instead of an object so that analysis can be done on the user ratings. This can be used to enhance the user experience.
- There are 39% of orders where there are no ratings. The customers need to be incentivized to give feedback on the orders so that meaningful data can be obtained.
- Four restaurants with ratings count more than 50 and average rating more than 4 are present in the data. These restaurants are performing well and can be given promotional offers.
- The customers can be incentivized by promotional offers for more number of orders. Currently, not many customers order food repeatedly.

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# Business Problem Overview

Problem Overview:

FoodHub is a food aggregator company with a smartphone app that allows online food delivery to customers in New York. FoodHub earns money by charging a fixed margin on each delivery order. The FoodHub app

- Allows food orders creation and receives restaurant confirmation
- Assigns a delivery person
- Provides maps to reach the restaurant and the customer
- Confirm the food drop off and accept ratings for the orders from the customer

FoodHub wants to analyze the orders data. The objectives are:

- to get an idea about demand of different restaurants
- enhance the customer experience and
- to improve business.

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# Solution approach

Data analysis:

FoodHub has Orders data available to analyze. The suggested approach to meet the objectives is as follows:

- Analyze the orders data to get information about the demand and revenue. This includes
  - number and types of restaurants, number of customers, frequency of orders
  - Most popular restaurants and cuisines and the days on which they are ordered most
  - revenue generation for different restaurants and food types
- Find  any strong correlations between the orders, ratings and revenue generated
- Find any problems with the data
- Based on the data analysis, suggest ways how to-
  - Make the customers happier resulting in higher ratings
  - Make the business stronger resulting in higher revenue

***Note**: You can use more than one slide if needed*

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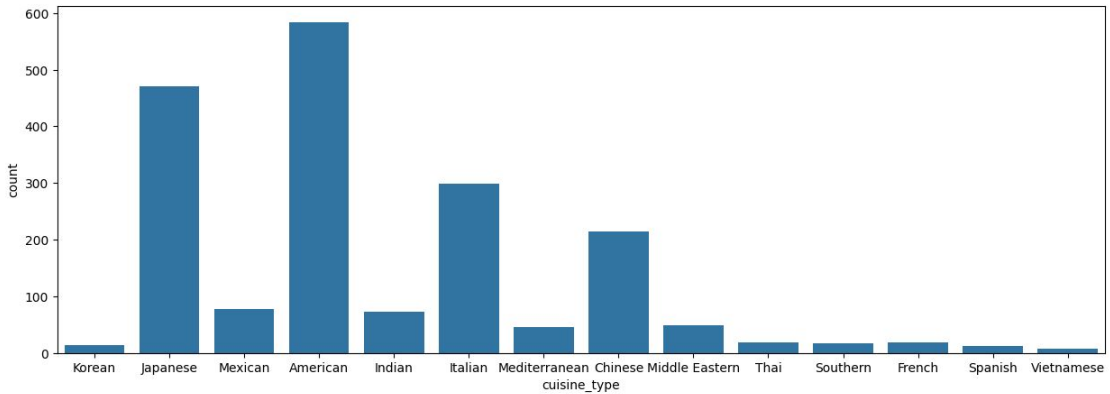
# Data Overview

- The orders data has 1898 orders in total.
- Orders data has 9 columns with following data types:
    - Int64 - order_id, customer_id, food_preparation_time, delivery_time
    - Object - restaurant_name, cuisine_type, day_of_the_week, rating
    - Float64 - cost_of_the_order
- There are no missing values in the data.
- The food preparation time varies between 20 min minimum  to 35 min maximum with and average of 27 min.
- There are a total of 736 (39%) orders that are not rated.

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# Univariate Analysis

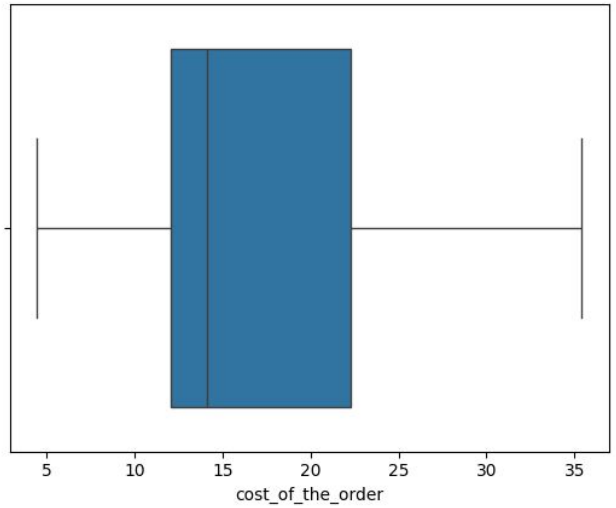
- The Orders data consists of 1898 total orders, 1200 customers and 178 restaurants.
- There are 14 different types of cuisines served in these restaurants.
  - Bulk of the orders are American, Japanese, Italian and Chinese cuisines..
  - Indian, Mexican, Middle eastern and Mediterranean cuisines are next in number of orders.
  - Thai, Southern, French, Spanish, Vietnamese and Korean cuisines have less number of orders.

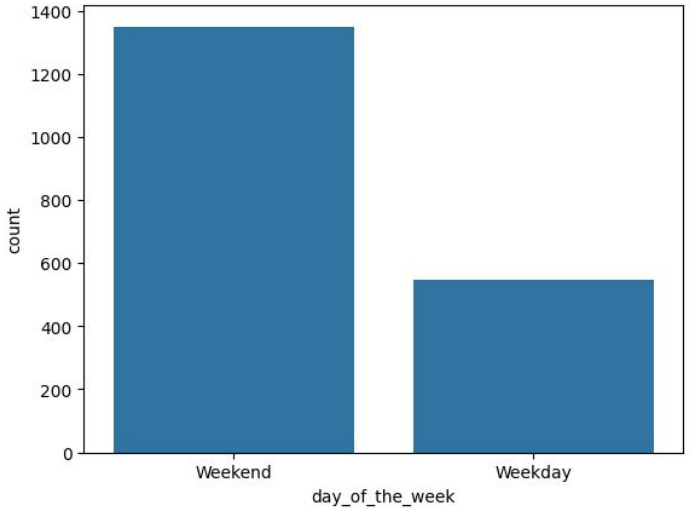



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# Univariate Analysis

- The cost of the orders ranges from $4 to $35 with and average of $16. The orders seem to be small individual orders and not big party orders. The cost data is a little bit skewed to the right but with no outliers.
- More than ⅔ of the orders are on weekends and less than ⅓ are on weekdays.

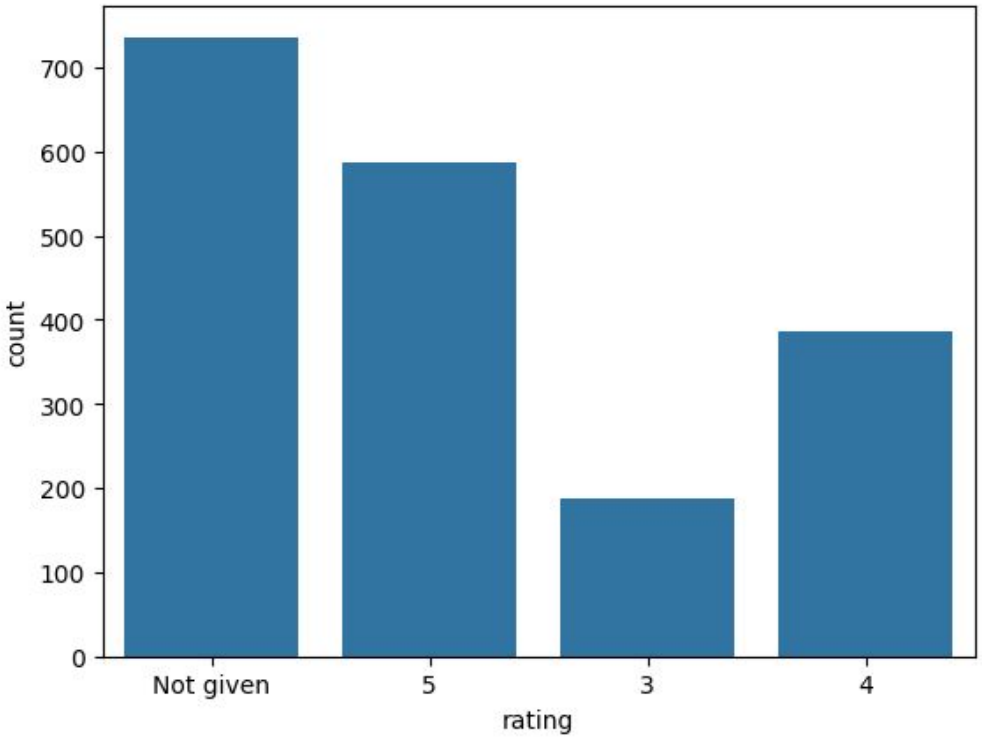



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# Univariate Analysis : Ratings

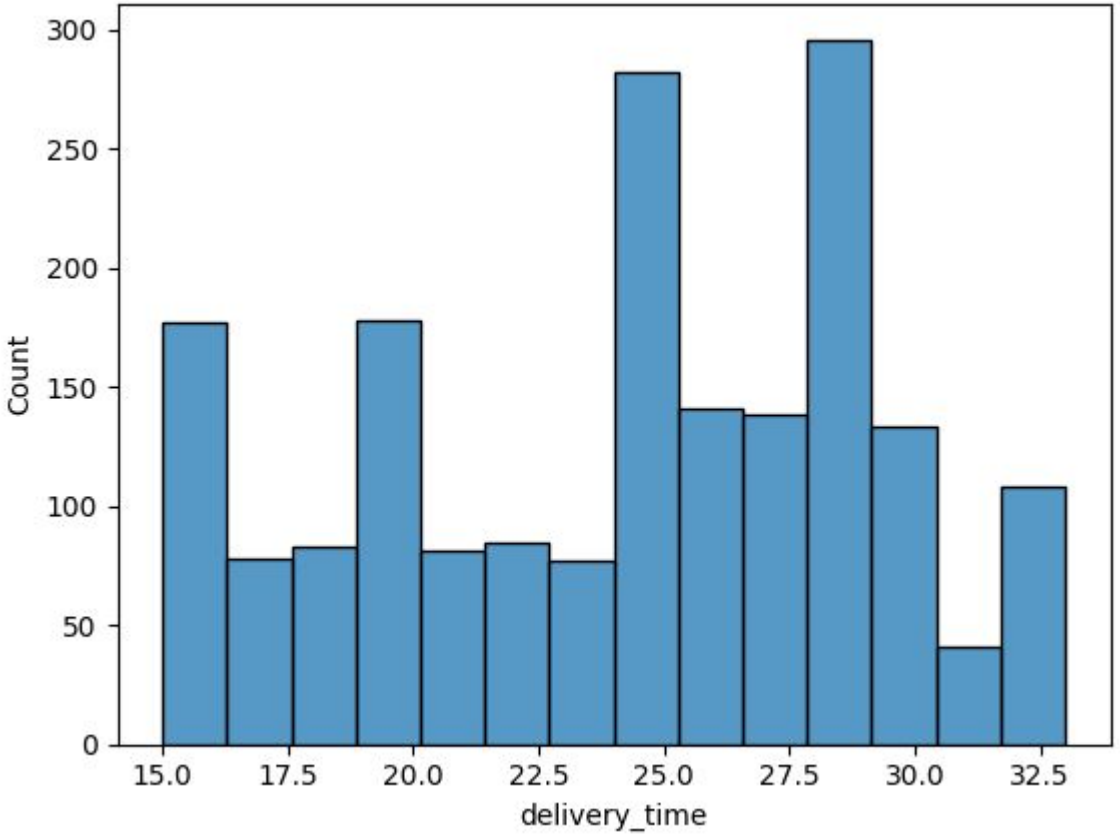
- About 39% or orders do not have any ratings. This seems to be a big percentage of orders with missing ratings.
- The order ratings are 5, 4 or 3 stars with approximately 3:2:1 ratio.
- The ratings column in the data is an object or a string. This prevents from any numerical analysis that could be done on the ratings.




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# Univariate analysis

Delivery time - Delivery times are between 15 to 33 min with average at 24 min.
There are no outliers.




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# Univariate analysis

One third of all orders are bagged by only 5 restaurants.

| restaurant_name | count |
|---|---|
| Shake Shack | 219 |
| The Meatball Shop | 132 |
| Blue Ribbon Sushi | 119 |
| Blue Ribbon Fried Chicken | 96 |
| Parm | 68 |

Most popular cuisine on the weekend: American

| cuisine_type | count |
|---|---|
| American | 415 |
| Japanese | 335 |
| Italian | 207 |
| Chinese | 163 |
| Mexican | 53 |

```
The number of total orders that cost above 20 dollars is: 555
Percentage of orders above 20 dollars: 29.24 %
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# Univariate analysis

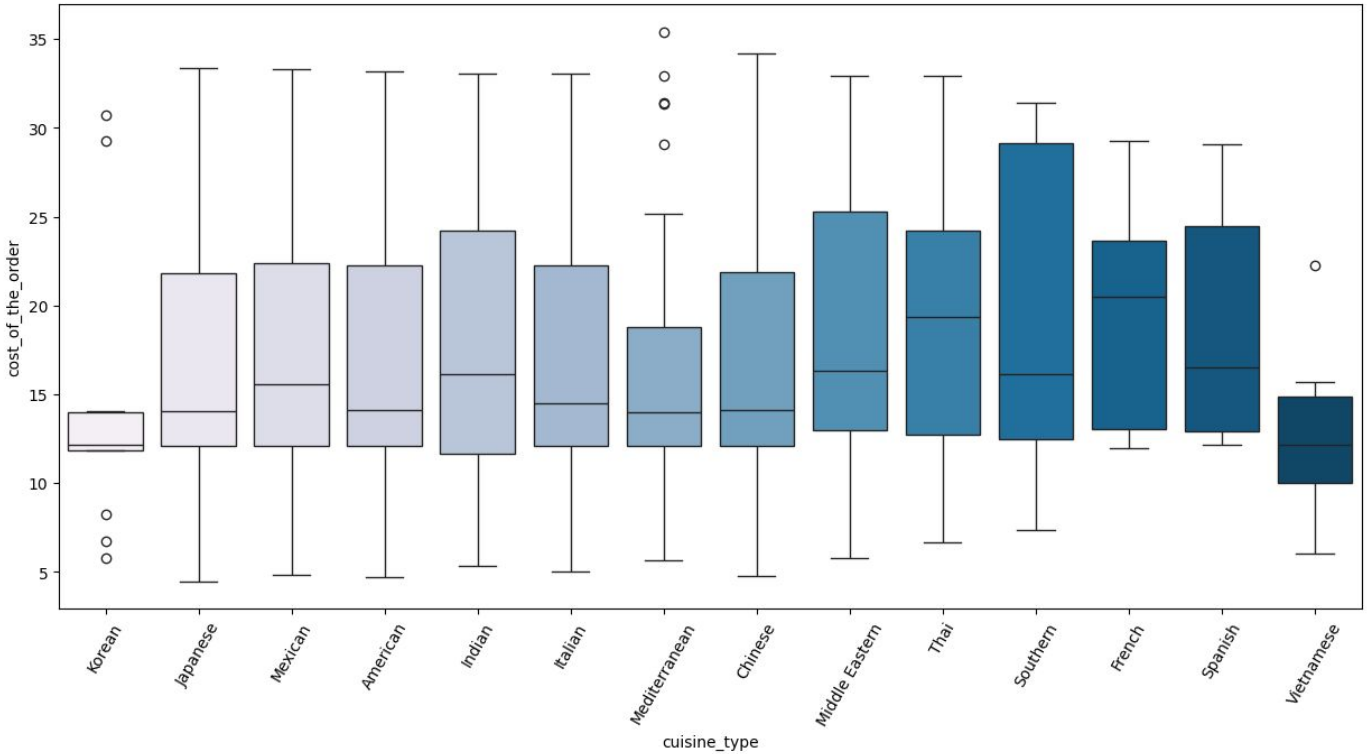
FoodHub’s top 5 frequent customers and their order counts are as shown below:

| customer_id | order_id |
|---|---|
| 52832 | 13 |
| 47440 | 10 |
| 83287 | 9 |
| 250494 | 8 |
| 276192 | 7 |

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# Multivariate Analysis- Cost of order Vs Cuisine

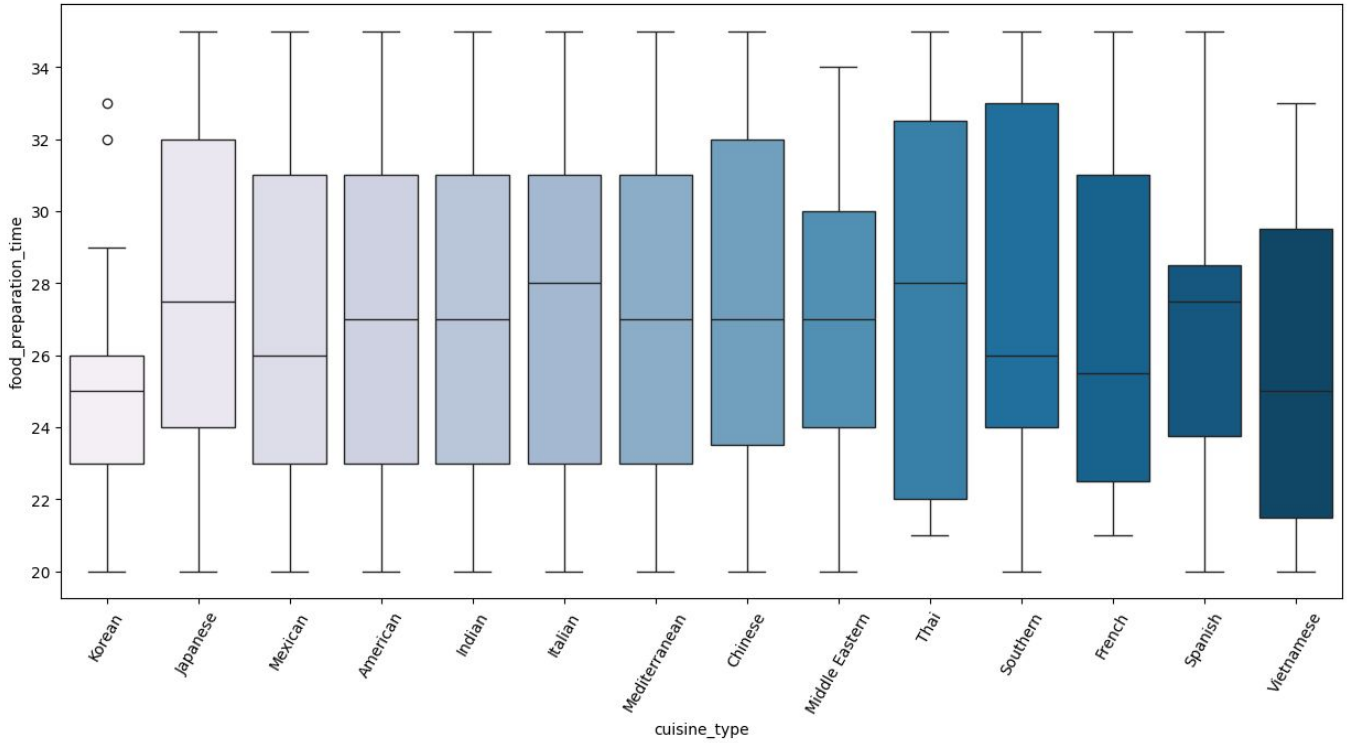
- Average of French cuisine order cost is highest
- Average of Vietnamese cuisine order cost is the lowest




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# Multivariate Analysis- Food preparation time Vs Cuisine type

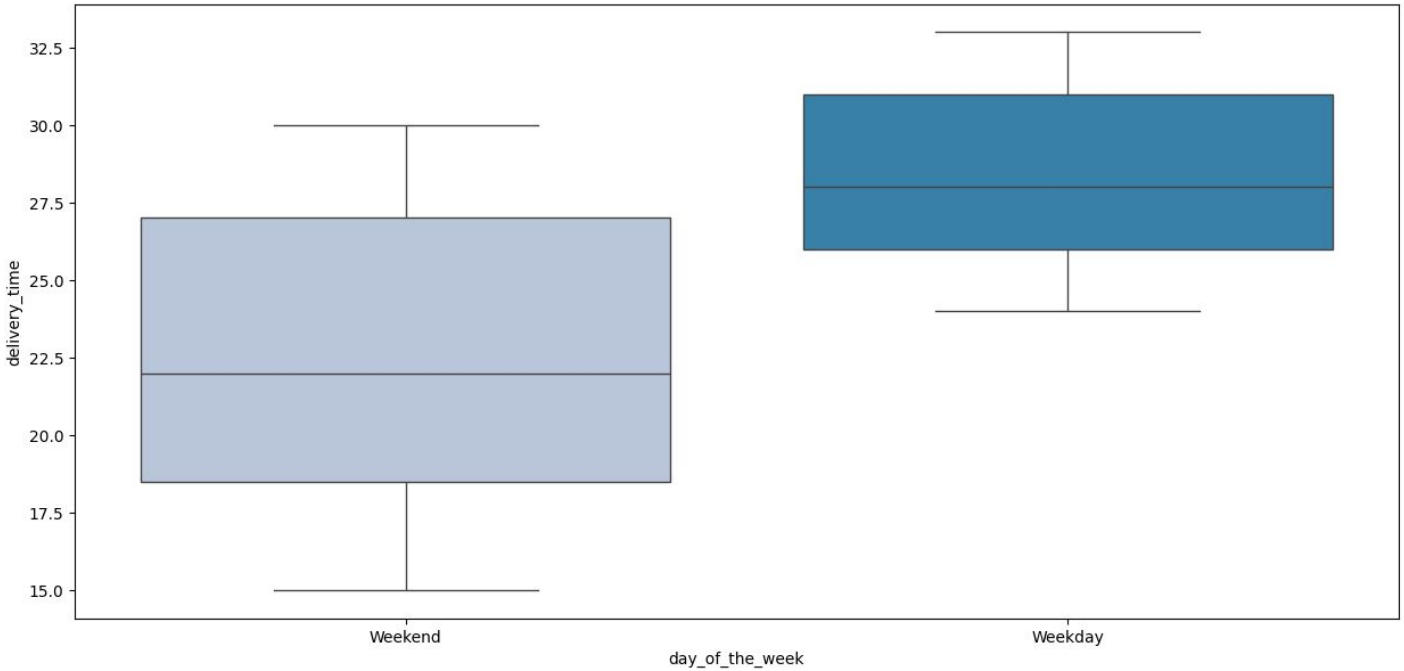
Except for Korean food, all other cuisine types have stable food preparation times with no outliers.




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# Multivariate analysis- delivery times Vs day of the week

Delivery times are much lower on the weekends even with twice as many orders as the weekdays!




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# Multivariate analysis

## Revenue generated by restaurants

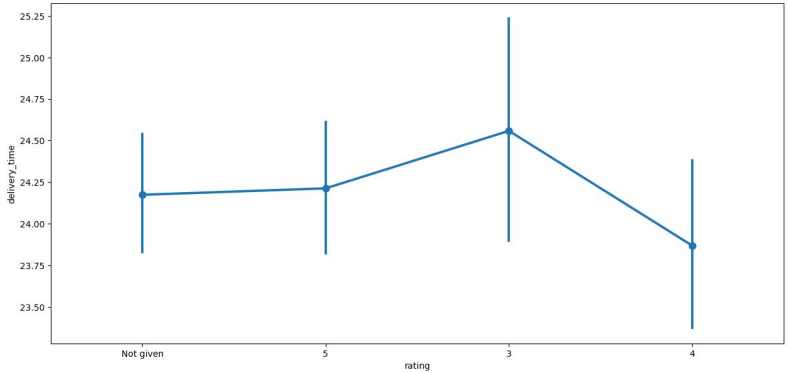
Here are our highest revenue generating restaurants.

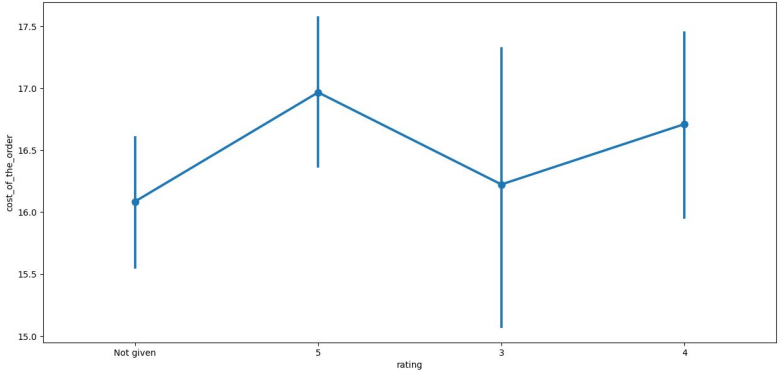
| | |
|---|---|
| **Shake Shack** | 3579.53 |
| **The Meatball Shop** | 2145.21 |
| **Blue Ribbon Sushi** | 1903.95 |
| **Blue Ribbon Fried Chicken** | 1662.29 |
| **Parm** | 1112.76 |
| **RedFarm Broadway** | 965.13 |
| **RedFarm Hudson** | 921.21 |
| **TAO** | 834.50 |
| **Han Dynasty** | 755.29 |
| **Blue Ribbon Sushi Bar & Grill** | 666.62 |
| **Rubirosa** | 660.45 |
| **Sushi of Gari 46** | 640.87 |
| **Nobu Next Door** | 623.67 |
| **Five Guys Burgers and Fries** | 506.47 |

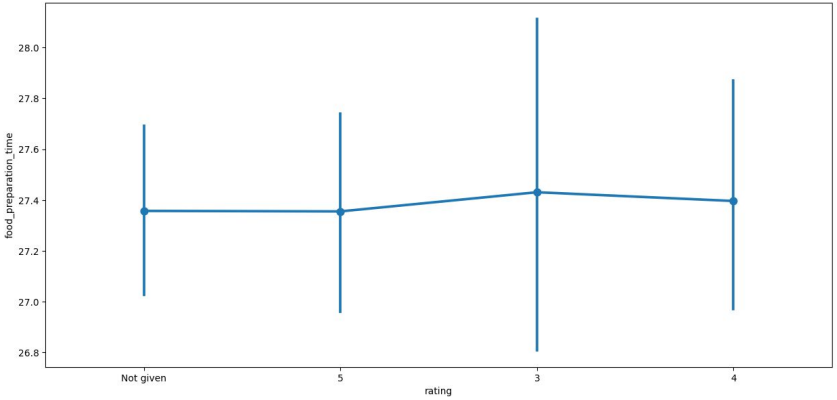
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# Multivariate analysis - How are ratings given?

The higher delivery time and lower cost of food is associated with a slightly lower rating given by users. But overall the rating associations are pretty flat.



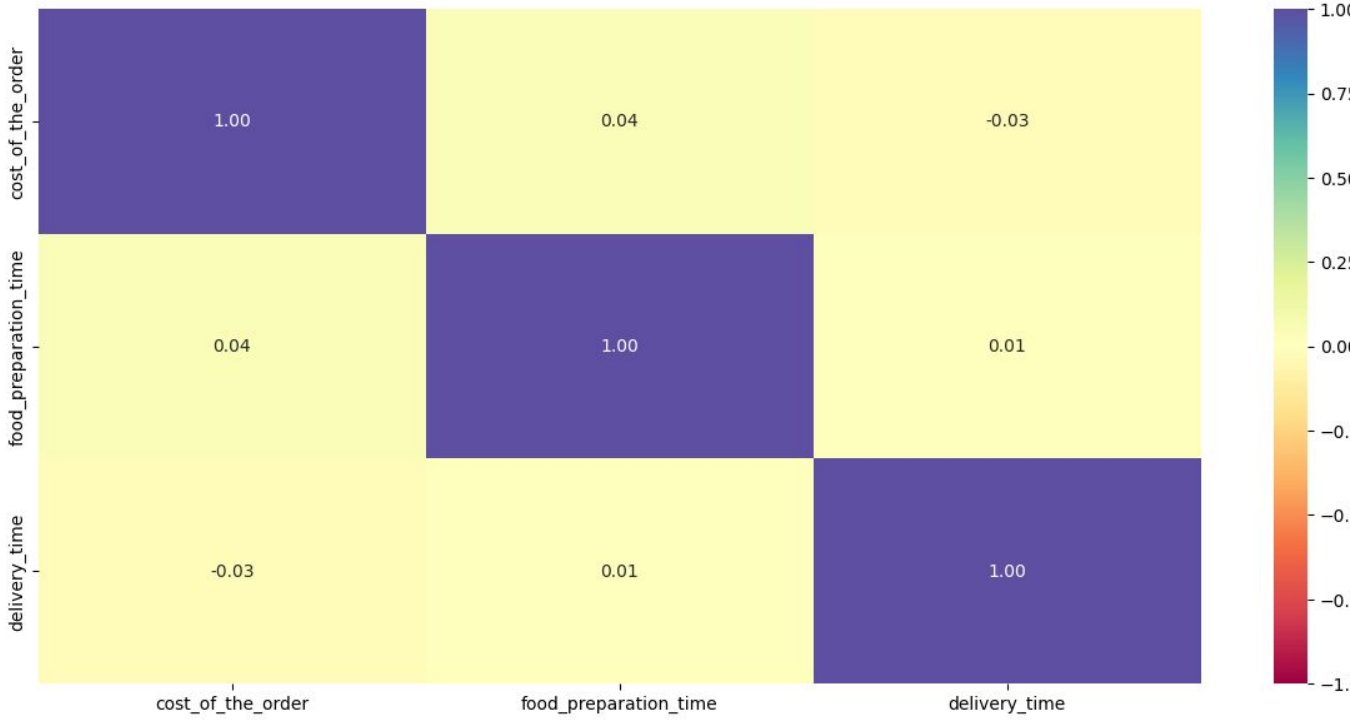




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# Multivariate analysis- Correlation between variables

There is no correlation between cost of the order, food preparation time and delivery time.




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# Restaurants eligible for promotional offers

The restaurants with more than 50 rating count and average rating more than 4 may be eligible for the promotional offers. We found these four!

| index | restaurant_name | rating |
|---|---|---|
| 0 | The Meatball Shop | 4.511904761904762 |
| 1 | Blue Ribbon Fried Chicken | 4.328125 |
| 2 | Shake Shack | 4.2781954887218046 |
| 3 | Blue Ribbon Sushi | 4.219178082191781 |

| index | restaurant_name | rating |
|---|---|---|
| 0 | Shake Shack | 133 |
| 1 | The Meatball Shop | 84 |
| 2 | Blue Ribbon Sushi | 73 |
| 3 | Blue Ribbon Fried Chicken | 64 |

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# Net Revenue

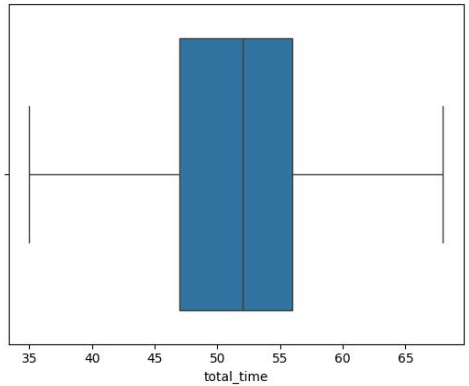
The company charges the restaurant 25% on the orders having cost greater than 20 dollars and 15% on the orders having cost greater than 5 dollars.

net revenue generated by the company across all orders is around- $6166.3

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# Total time required to get the food and mean delivery time

The number of total orders that took more than 1 hour is: 200
Percentage of total orders that took more than 1 hour is: 10.54 %




The mean delivery time on weekdays is around 28 minutes
The mean delivery time on weekends is around 22 minutes

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